

P&ID Redline Punch List & BOM Review

Hayfield Ridge Amine Facility — MPLX (Kinder Morgan Treating LP) · Lea County, New Mexico

Report date: 2026-08-09 **Source:** Field-redlined P&ID set, 40 sheets (HAY-PID-01001 ... HAY-PID-08005)
Prepared for: Construction / Commissioning team **Basis:** Every cloud, note, highlight & strikethrough transcribed; two-pass verification

1 Executive Summary

40 P&IDS REVIEWED	39 SHEETS WITH REDLINES	430 PUNCH ITEMS	86 CRITICAL ITEMS	72 BOM LINE ITEMS	18 ENGINEERING QUESTIONS
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The complete 40-sheet redlined P&ID set for the Hayfield Ridge Amine Facility was reviewed sheet-by-sheet at high magnification. Every red mark, cloud, strikethrough, and pink/magenta highlight was transcribed and converted to a punch item. One sheet is clean (HAY-PID-01002 Gas Cooler — printed “FUTURE SCOPE”, no marks); the other 39 sheets carry field markups.

Highest-risk findings (act first):

- **Missing relief devices.** PSVs / relief assemblies are highlighted “Missing” at: V-22001 gas scrubber (1435 psig service), S-1100 inlet coalescing filter (PSV-1100, 1440 psig), E-2 reboiler thermal PSV, AC-1 amine cooler (PSV AC-1), all four VCU knock-out drums (PSV-301/-401/-302/-402), air system PSV-7002, generator fuel PSV-7006, plus missing PSV tailpipes at T-1 and F-71001. **No pressure test / startup until resolved.**
- **Overpressure routing question (HP pumps).** Field notes PSV P-3C (set 1440 psig) relieves via 3"-V-0073-A2 to a 165# MAWP vessel — engineering must verify relief path adequacy.
- **Check valve installed backwards** on 2"-DR-7001-F2AS (Blowcase System).
- **Instrument air cross-connected to the methanol inlet port** on the JT Fuel Skid (connection E) — IA must be relocated and methanol connected.
- **Hot Oil system gaps not on ISOs:** FCV-45001A control-valve station and PCV-2001 blanket-gas regulator “don’t exist / not on ISO’s”; burner vent valves XV-20003A / XV-20006A missing.
- **Safety instrumentation missing:** level switch stacks (LSHH/LSH/LSL/LSLL) on all four KO drums; ESD/SDV wiring and tubing incomplete across the plant; slop-tank grounding station missing.

Schedule impact. The dominant themes are (1) relief-system completion, (2) instrument installation/wiring/tubing (largest item count — I&E), (3) small-bore valve additions (~60 valves ½"-3"), and (4) pipe supports/shoes/U-bolts. Long-lead risks: PSVs and the 14" valves (2 off) on the VCU vent headers, FCV-45001A (8" control valve), static mixer X-2, and orifice plates with specified bores. Several “SIZE HOLD” PSV vents remain unreleased by engineering — these block procurement. Underground (U/G) tie-ins noted on multiple headers should be completed before paving/backfill close-out.

Numbering: items are keyed <sheet>-<nn> (e.g. 01001-07). Verbatim field text is shown in *red italics*. Priorities: **CRITICAL** pre-pressure-test / startup / safety · **HIGH** before mechanical completion · **MEDIUM** before turnover · **LOW** documentation/cosmetic.

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01001 Slug Catcher · G	02015A Exhaust Stack (Hot Oil) · E
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02001 Inlet Coalescing Filter · F	02015D Combustion Air Blower · E
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02004 Amine Flash Tank · E	03003 Glycol Region · F
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02011 HP Amine Pumps · G	07005 Methanol Tank · C
02012 Reboiler Skid · F	07006 Generator · D
02013 T.O. 400 GPM Train · F	08001 Amine Makeup Tank · E
02014 BTEX Dehy Unit-I · F	08002 TEG Glycol Makeup Tank · E
02115 T.O. 640 GPM Train · F	08003 Chemetherm Makeup Tank · E

3 Master Punch List — by P&ID

HAY-PID-01001 Slug Catcher (X-11001 / X-12001)

Rev G

43 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
01001-01	2"-IA-7001-L2	PCV-1001/-1002	"Install (Inlet/outlet)" — install the two 1" IA take-off valves/drops feeding PCV-1001 & PCV-1002.	Piping	HIGH
01001-02	Slug catcher area	X-11001/X-12001	"Install Platforms (2)" — install two access platforms.	Structural	MED
01001-03	Inlet gas	PIT-1001	"Wire" — pull/terminate instrument wiring.	Electrical	HIGH
01001-04	16"-GH-0001-F2AS	16" CSO valve	"Rotate Handle Due South" — reorient valve handle.	Piping	MED
01001-05	16"-GH-0001-F2AS	Inlet spec blind	"Roll after PSSR" — red-drawn spectacle blind at inlet 16" CSO valve; roll to open position after PSSR.	Operations	CRITICAL
01001-06	16"-GH-0002-F2AS	Outlet spec blind	"Roll after PSSR" — second red-drawn spectacle blind on outlet gas header to 16" launcher; roll after PSSR.	Operations	CRITICAL
01001-07	16"-GH-0002-F2AS	—	Red-drawn 1" valve w/ plug teed off outlet header at the blind — install.	Piping	HIGH
01001-08	16"-GH-0002-F2AS	—	"U/G" at OUTLET GAS HEADER TO 16" LAUNCHER flag — record underground transition (as-built).	Engineering	LOW
01001-09	16"-GH-0002-F2AS	—	"U/G" break-arc right of ESD station — record underground segment (as-built).	Engineering	LOW
01001-10	16"-GH-0001-F2AS	SV-1001, ZSO/ZSC-1001	"Wire + Tubing" — complete wiring and pneumatic tubing at inlet ESD valve station.	I&E	CRITICAL
01001-11	16"-GH-0002-F2AS	SV-1002, ZSO/ZSC-1002	"Wire + Tubing" — complete wiring and tubing at outlet ESD valve station.	I&E	CRITICAL
01001-12	Inlet PSV vent	—	Red-drawn ½" valve w/ cap labeled "½"" on vent tailpipe below inlet PSV — install.	Piping	HIGH
01001-13	16"-GH-0001-F2AS	—	Red-drawn small-bore valve + cap on stub near 16"x10" control valve cloud — install.	Piping	HIGH
01001-14	3"-GH-0015-A2AS	PSV-1001	"Confirm" — verify relief set pressure 1430 PSIG (fire) against nameplate/certs.	Mechanical	CRITICAL
01001-15	3"-GH-0017-A2AS	PSV-1003	"Confirm" — verify relief set pressure (X-12001) against nameplate/certs.	Mechanical	CRITICAL
01001-16	16"-GH-0003-F2AS	—	Red-drawn valve + blind flange on stub at elbow — install.	Piping	HIGH
01001-17	16"-GH-0006-F2AS	—	Red-drawn vertical valve + cap teed up off line — install.	Piping	HIGH
01001-18	Outlet gas	PIT-1002	"Wire" — pull/terminate instrument wiring.	Electrical	HIGH
01001-19	X-11001 level bridle	LIT-1001	"4x2 / Remove" + strike on ½" valve between LIT-1001 and control override — remove valve; verify 4"x2" reducer requirement.	I&E	HIGH
01001-20	X-12001 level bridle	LIT-1002	Same markup as 01001-19 at X-12001 — remove ½" valve; verify 4"x2" reducer.	I&E	HIGH
01001-21	X-11001 bridle	LS-1001	Red "2"" (twice) at bridle isolation valves — verify/upsized isolation valves to 2".	Piping	MED
01001-22	X-12001 bridle	LS-1002	Red "2"" (twice) — same verification at X-12001 bridle.	Piping	MED
01001-23	4"-LH-0002-F2AS-1½"H-ET	X-11001	Red-drawn drain valve + plug on stub — install.	Piping	HIGH
01001-24	4"-LH-0005-F2AS-1½"H-ET	X-12001	Red-drawn drain valve + plug on stub — install.	Piping	HIGH
01001-25	4"-LH-0003-F2AS-1½"H-ET	LV-1001 bypass	"Missing" (x2, pink highlight) — 4" cone strainer and 4" FD ("SIZE HOLD") on NC bypass not installed. Install; engineering to release size hold.	Piping	CRITICAL

01001-26	4"-LH-0006-F2AS-1½"H-ET	LV-1002 bypass	"Missing" (×2) — same missing 4" cone strainer + FD (SIZE HOLD) at X-12001 letdown bypass.	Piping	CRITICAL
01001-27	Letdown loop	LY/LV-1001	"Wire" — complete loop wiring.	I&E	HIGH
01001-28	Letdown loop	LY/LV-1002, LSH/LSLL-1002	"Wire" — complete loop wiring incl. level safety switches.	I&E	CRITICAL
01001-29	Signal runs, X-11001	—	Red slashes across dashed signal lines + "Wire" — pull signal wiring.	Electrical	HIGH
01001-30	4"-LH-0004-F2AS-1½"H-ET	—	"Install pipe shoes (3)" — install three pipe shoes.	Piping	HIGH
01001-31	Vessel saddles	X-11001/X-12001	"Grout + Ground" — grout saddles and install grounding.	Civil / Elec	HIGH
01001-32	IA header drops	SC#1 / SC#2	"Add valves (2) - SC#1 + SC#2" — add one IA isolation valve per slug catcher.	Piping	HIGH
01001-33	12"-GH-0010-F2AS	PT-1001A	Red cloud + "Wire" + leader re-tapping PT-1001A onto the 12" line — relocate tap, wire transmitter.	I&E	HIGH
01001-34	12"-GH-0010-F2AS	FE/FT-1001, TW/TE-1001	Large hatched cloud over meter-run instruments + sketch of orifice run: "Installing Orifice Meter or RO + FIT ?" — engineering to confirm metering design; install per answer. (Eng. Question Q1)	Engineering	HIGH
01001-35	FV-1001 cloud	FV/FY-1001	"Add" — red-drawn valves: 2" (×2) + ½" (×1) — install.	Piping	HIGH
01001-36	Left of FV cloud	—	"U/G" break-arcs — record underground segments (as-built).	Engineering	LOW
01001-37	3"-LH-0011-F2AS-1½"H-ET	—	Red "2"-LH-0009-F2AS-1H-ET" added above unlabeled line; flag corrected to "Coalescing Filter / HAY-PID-02101"; "Δ 3x2" at branch — correct line tags on drawing; verify 3"x2" reducer at branch.	Engineering	MED
01001-38	2"-DR-7003-F2AS	—	Red-drawn new line + off-page flag to Liquid Blowcase (HAY-PID-07004) — add line to drawing; verify field tie-in (matches 07004 sheet markup).	Engineering	MED
01001-39	Adjacent riser	PT-1001B	Red cloud + curved arrow — relocate PT-1001B tap to adjacent riser.	I&E	HIGH
01001-40	1"-GH-3006-F2AS	—	Red/pink highlight along line to Dehy Train I & II + "U/G" — record underground routing (as-built).	Engineering	LOW
01001-41	Blowdown drain, X-12001	—	"Wire + Tubing" — complete wiring/tubing in blowdown drain area.	I&E	HIGH
01001-42	4" FC valve	ZS0/ZSC-1007	"wire" — wire position switches on ESD blowdown valve.	I&E	CRITICAL
01001-43	Near 4" FC station	—	"U/G" + break-arc — record underground transition (as-built).	Engineering	LOW

HAY-PID-01003 Gas Scrubber (V-21001 / V-22001)

Rev E

23 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
01003-01	10"-GH-0021-F2AS	V-22001 inlet	"Missing U-bolt (2)" — install two 10" U-bolts at inlet ESD station.	Piping	HIGH
01003-02	10"-GH-0021-F2AS	—	"Secure 2 U-bolt / Missing 1 U-bolt" — secure two existing U-bolts, install one missing.	Piping	HIGH
01003-03	Inlet ESD valve	SV-1004	"Tubing" — complete pneumatic tubing at solenoid/quick-connect.	I&E	CRITICAL
01003-04	2"-GH-0022-F2AS	—	"Add Gate Valve" — install 2" gate valve in bypass leg of inlet station.	Piping	HIGH
01003-05	V-22001 top	—	"Add platform (AR)" — access platform as required.	Structural	MED
01003-06	3"-GH-0031-A2AS vent	V-22001 PSV	"Missing" + pink highlight over ENTIRE PSV package (PSV, 3" vent to ATM, 1½"-GH-0030-F2AS inlet, ½" bleed) — supply & install complete relief assembly (set per V-22001, 1435 PSIG design).	Mechanical	CRITICAL
01003-07	2"-GH-0027-F2AS / 10"-GH-0026-F2AS	—	Pink/red triangle on 1" vent valve at junction (to Coalescing Filter) — verify/install 1" valve.	Piping	HIGH

01003-08	V-22001 bridle	LIT-1004	Red <i>“LT”</i> circle + <i>“3X2”</i> capped-stub sketch — mount LIT-1004 on 3"x2" reducer with cap.	I&E	HIGH
01003-09	V-21001 bridle	LIT-1003 / LSH-1003	<i>“Remove”</i> + strike — remove ½" valve/plug between instruments.	I&E	HIGH
01003-10	V-21001 bridle	—	Red <i>“3”</i> (twice) at bridle isolation valves — verify/upsized to 3".	Piping	MED
01003-11	V-22001 bridle	LIT-1004	<i>“Remove”</i> + strike — remove ½" valve at V-22001 bridle.	I&E	HIGH
01003-12	V-22001 bridle	—	Red <i>“3”</i> (twice) — verify/upsized bridle isolation valves to 3".	Piping	MED
01003-13	V-22001 manway	MW	<i>“Torque”</i> — torque manway bolting to spec.	Mechanical	HIGH
01003-14	V-22001 skirt	—	<i>“Ground”</i> — install vessel grounding.	Electrical	HIGH
01003-15	2"-LH-0010-F2AS boot drain	—	Red-drawn ½" valve + plug teed off 2" boot-drain stub — install.	Piping	HIGH
01003-16	2"-LH-0009-F2AS area	Bridle drain	<i>“Missing 3" blind + ½" valve”</i> — install 3" blind flange (w/ gasket & bolting) and ½" valve at bridle drain below lower nozzle.	Piping	CRITICAL
01003-17	2"-LH-0009-F2AS	—	<i>“Shim pipe shoe”</i> — shim shoe downstream of V-22001 outlet valve.	Piping	HIGH
01003-18	2"-LH-0009-F2AS	ESD-1004	<i>“Missing pipe shoe + shim”</i> — install shoe + shim at FC letdown station.	Piping	HIGH
01003-19	Drain leg	—	<i>“lift pipe & put shoe on beam”</i> (1) — reset shoe onto beam.	Piping	HIGH
01003-20	2"-LH-0010-F2AS	2" FC (SIZE HOLD)	<i>“lift pipe & put shoe on beam”</i> (2) — reset shoe at letdown station.	Piping	HIGH
01003-21	ESD-1004	ZSO/ZSC-1004	<i>“Tubing + lower latching switch to grade (Tubing)”</i> — run tubing; relocate latching switch to grade.	I&E	CRITICAL
01003-22	2"-LH-0009-F2AS	—	Red note <i>“2"-LH-0009-F2AS / P&ID-02001”</i> + circled 1 — line continues to HAY-PID-02001; update drawing continuity (mate of 02001-14).	Engineering	LOW
01003-23	3"-LH-0011-F2AS-1"H-ET	—	Red <i>“3”</i> size prefix added; flag sheet number struck, <i>“02101”</i> written — correct line size & continuation sheet on drawing.	Engineering	LOW

HAY-PID-02001 Inlet Coalescing Filter S-1100 (400 GPM Amine Plant)

Rev F

16 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
02001-01	10"-GH-0028-F2AS	—	Red sketch: two tap valves w/ bleeds flanking orifice-flange pair + <i>“RO + FIT ?”</i> — engineering to confirm whether RO + flow transmitter go here; install per answer. (Eng. Question Q2)	Engineering	HIGH
02001-02	Filter inlet	FV S-1100	<i>“Remove Wood block”</i> — remove shipping block from 10" FC valve.	Operations	CRITICAL
02001-03	S-1100 relief	PSV-1100	<i>“Missing”</i> + pink highlight over full assembly (PSV set 1440 PSIG fire, 2" TO ATM tail, 2"x1" reducer, ½" bleed, 2" CSO block) — supply & install.	Mechanical	CRITICAL
02001-04	Meter Skid #1	Mustang SCS panel	<i>“Conduit Installed, Need to pull wire”</i> — pull & terminate wire to heated enclosure/stream splitter.	Electrical	HIGH
02001-05	Level controller	LC S-1100B	<i>“Rotate Due West”</i> — reorient controller.	I&E	MED
02001-06	Level controller	LC S-1100A	<i>“Rotate Due East”</i> — reorient controller.	I&E	MED
02001-07	2"-C-0201-F2AS-1.5"H-ET	LCV-1100B	<i>“Shoes”</i> — install two pipe shoes on 2" condensate lines at station.	Piping	HIGH
02001-08	Shutdown valve	SDV-1100, ZSO/ZSC-1100	<i>“Tubing”</i> — complete pneumatic tubing.	I&E	CRITICAL
02001-09	Below LC-1100A	—	<i>“Tubing”</i> — complete tubing on vertical 2" run/header connections.	I&E	HIGH

02001-10	LCV-1100A station	—	<i>“shoe”</i> — install one pipe shoe.	Piping	HIGH
02001-11	S-1100	—	<i>“Ground vessel + skid”</i> — install grounding.	Electrical	HIGH
02001-12	S-1100 skirt	—	<i>“Grout”</i> — grout vessel base.	Civil	HIGH
02001-13	Inlet riser u/s N1	—	Red <i>“½”</i> + arrowhead at coupling — install ½" valve/plug at open coupling.	Piping	HIGH
02001-14	2"-LH-0009-F2AS	—	Red <i>“2"-LH-0009-F2AS / P&ID-01003”</i> + tie of 2"-C-0201 drop into gas outlet header — update drawing; verify field tie-in (mate of 01003-22).	Engineering	MED
02001-15	2"-LH-0009-F2AS-1"H-ET	—	<i>“U/G”</i> + break — record underground segment of gas outlet header (as-built).	Engineering	LOW
02001-16	Outlet header flag	—	Sheet number struck, <i>“01001”</i> written — correct continuation flag.	Engineering	LOW

HAY-PID-02002 Amine Contactor T-1 (400 GPM Amine Plant) Rev E10 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
02002-01	T-1 relief	PSV T-1	<i>“Confirm”</i> — verify set pressure 1440 PSIG (fire) against nameplate.	Mechanical	CRITICAL
02002-02	2"-V-0033-A2	PSV vent	<i>“Missing”</i> + pink highlight — install 2" PSV vent tailpipe to ATM.	Piping	CRITICAL
02002-03	Same tailpipe	—	<i>“Add Brace”</i> — brace/support the vent tailpipe.	Piping	HIGH
02002-04	T-1 column	PDT T-1	<i>“Missing”</i> + pink highlight along full ½" SS tubing run (top to bottom tap) — install dP impulse tubing (~55 ft + fittings).	I&E	HIGH
02002-05	6"-LA-0025-F2	—	<i>“Secure U-bolt”</i> — secure U-bolt on lean amine line near N3.	Piping	HIGH
02002-06	T-1 manway M1 (18")	—	<i>“Torque after Inspection”</i> — torque lower manway after internal inspection.	Mechanical	HIGH
02002-07	8"-G-0103-F2AS	8" CV60N	<i>“Rotate 90° so Valve is Horizontal”</i> — reorient inlet valve.	Piping	MED
02002-08	Lower head, near N9	—	<i>“Remove”</i> + X — remove 2" capped valve/stub.	Piping	HIGH
02002-09	T-1 skirt	—	<i>“Grout”</i> — grout baseplate.	Civil	HIGH
02002-10	6"-RA-0009-F2AS-SR	R0 T-1	<i>“Confirm”</i> — verify restriction orifice bore 1.5" per datasheet.	I&E	HIGH

HAY-PID-02003 Outlet Scrubber V-1 (400 GPM Amine Plant) Rev E7 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
02003-01	V-1 relief	PSV V-1	<i>“Confirm”</i> — verify set pressure 1440 PSIG (fire, 1D2 orifice) against nameplate.	Mechanical	CRITICAL
02003-02	PSV V-1 tailpipe	—	<i>“Add Brace”</i> — brace 2" atmospheric vent riser.	Piping	HIGH
02003-03	V-1 nozzle N3 (2")	PI V-1	<i>“Remove”</i> + X — remove small gauge/bleed valve at N3 between PI isolation and vessel.	I&E	HIGH
02003-04	V-1 manway M1 (18")	—	<i>“Torque”</i> — torque manway cover.	Mechanical	HIGH
02003-05	V-1 skirt	—	<i>“Grout”</i> — grout vessel base.	Civil	HIGH
02003-06	LV V-1 area	—	<i>“Install platform”</i> — access platform for level valve/instrument area.	Structural	MED
02003-07	1"-C-0008-F2AS-SR	R0 V-1	<i>“Missing”</i> + pink highlight over printed “CHOKE NIPPLE / 3/8" BORE” — install 1" choke nipple, 3/8" bore, on condensate to flash tank.	Piping	CRITICAL

HAY-PID-02003A Gas Cooler AC-4 (400 GPM Amine Plant) Rev E1 item

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
02003A-01	Louver actuator run	TY AC-4	<i>“Tubing”</i> — install pneumatic tubing from TY AC-4 to automated louver actuator.	I&E	HIGH

HAY-PID-02004 Amine Flash Tank V-2 (400 GPM Amine Plant) Rev E12 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
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02004-01	8"-RA-0010-F2AS-SR	—	Red-drawn ½" valve + cap labeled <i>"½"</i> on top of rich amine run — install high-point vent.	Piping	HIGH
02004-02	V-2 relief	PSV V-2	<i>"Confirm"</i> — verify set pressure 150 PSIG (6R8) against nameplate.	Mechanical	CRITICAL
02004-03	PCV V-2B station	u/s bleed	Red <i>"1"</i> + cap mark — change upstream bleed valve to 1", cap outlet.	Piping	HIGH
02004-04	PCV V-2B station	d/s bleed	Printed ¾" struck, <i>"1"</i> + cap — change downstream bleed to 1", cap outlet.	Piping	HIGH
02004-05	V-2 nozzle M4 (4")	—	Small red check at flange pair (KMTD/MPLX break) — field verification mark; confirm intent/torque status.	Mechanical	MED
02004-06	LG V-2A bridle, N8	2" GVTONT CSO	Red <i>"CB10"</i> — change upper bridle valve spec GVTONT → CB10 (retain CSO).	Piping	HIGH
02004-07	LG V-2A bridle, N9	2" GVTONT CSO	Red <i>"CB10"</i> — change lower bridle valve to CB10.	Piping	HIGH
02004-08	LIT V-2 standpipe	LIT V-2	<i>"Secure Wire"</i> (two arrows) — secure wire/heat-trace along bridle standpipe.	Electrical	HIGH
02004-09	LIT V-2 bridle	LIT V-2	<i>"Wire"</i> — complete/terminate wiring at transmitter.	Electrical	HIGH
02004-10	1"-DR-0003-A2AS	—	<i>"Remove By MPLX"</i> + X — remove flanged drain valve adjacent saddle.	Piping	HIGH
02004-11	V-2 saddles	—	<i>"Grout (2)"</i> — grout both saddles.	Civil	HIGH
02004-12	V-2 level end	—	<i>"Install Platform"</i> — access platform at level-instrument end.	Structural	MED

HAY-PID-02005 Amine Filters F-1 / F-2 (400 GPM Amine Plant) Rev E14 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
02005-01	2"-V-0060-A2AS	Skid edge	<i>"Add 2" valve / (CSO)"</i> — install 2" CSO valve at skid-edge vent to drain system.	Piping	HIGH
02005-02	F-1 relief	PSV F-1	<i>"Confirm"</i> — verify set 150 PSIG (fire, 1½F2).	Mechanical	CRITICAL
02005-03	F-1 degasser	—	<i>"Missing ½" Tubing"</i> + pink highlight — install ½" SS tubing run (check valve to ½"-V-0060-A1 tie-in).	I&E	HIGH
02005-04	Skid #1 limit (west)	—	Red-drawn 2" valve w/ X at skid boundary above rich-amine entry — pictorial for 02005-05; install with it.	Piping	HIGH
02005-05	6"-RA-0012-A2-1.5"H-SR	—	<i>"Add 2" check + Block"</i> — install 2" check valve + 2" block valve at scope break (flush/aux connection).	Piping	HIGH
02005-06	F-1 inlet meter	FE F-1	<i>"Confirm"</i> — verify orifice bore 2.0".	I&E	HIGH
02005-07	F-1 meter run	—	Red-drawn ¾" valve + plug below FE F-1 — install vent/bleed.	Piping	HIGH
02005-08	6"-RA-0013-A2-1.5"H-SR	—	Red-drawn ¾" valve + plug near F-1 outlet riser — install drain/vent.	Piping	HIGH
02005-09	F-2 degasser	—	<i>"Tighten ½" Tubing Ferrule"</i> — tighten compression fitting.	I&E	HIGH
02005-10	1"-V-0071-A1	PSV F-2 tail	<i>"Doesn't Exist."</i> + X on printed 1" CSO valve — valve shown on drawing is not in field.	Piping	CRITICAL
02005-11	1"-V-0071-A1	—	<i>"Needs to Be Added Inside Skid limits."</i> — install the missing 1" CSO valve + tailpipe work inside skid.	Piping	CRITICAL
02005-12	6" outlet header	—	Red-drawn valve + plug (size unmarked, ~¾") on header toward L/R exchanger — install low-point drain.	Piping	HIGH
02005-13	2"-DR-0100-A1/A2	—	Red continuation flag <i>"2-DR-0100-A2 / HAY-PID-02016"</i> at west end of drain header — correct drawing continuity; verify field tie-in.	Engineering	MED
02005-14	2"-DR-0100-A1 (east)	—	Red emphasis chevrons on printed drain flag — verify/connect continuation to HAY-PID-02006.	Piping	MED

HAY-PID-02006 L/R Exchanger E-1 (400 GPM Amine Plant) Rev E4 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
02006-01	6"-LA-0022-A1	TI/TW E-1D	Red cloud + tag <i>"E-1D"</i> on unlabeled TI/TW bubble — install & tag temperature indicator with thermowell on E-1 lean amine outlet.	I&E	HIGH
02006-02	6" crossover branch	—	Red-drawn valve + plug (no size) upstream of 6" block valve — install vent/drain bleed.	Piping	HIGH

02006-03	2"-DR-0100-A2 (west)	—	Red <i>"2-DR-0100-A2"</i> filled into blank continuation flag over HAY-PID-02005 — correct drawing (mate of 02005-13).	Engineering	LOW
02006-04	2"-DR-0100-A1 (east)	—	Red emphasis on printed flag to HAY-PID-02007 — verify continuation/tie-in.	Piping	MED

HAY-PID-02007 Amine Booster Pumps / Lean Solids Filter (P-1A/B, F-4)Rev E8 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
02007-01	6"-LA-0018-A1-1.5"H	P-1A discharge	Red <i>"½"</i> + drawn valve between check and block — install ½" valve (vent/gauge).	Piping	HIGH
02007-02	6"-LA-0019-A1-1.5"H	P-1B discharge	Red <i>"½"</i> + drawn valve — install ½" valve.	Piping	HIGH
02007-03	1"-V-0076-A1	1" C1203F LSD	<i>"Doesn't Exist."</i> [sic] + cloud — printed 1" shutdown valve not in field; reconcile install vs drawing revision. (Eng. Question Q13)	Engineering	HIGH
02007-04	1"-V-0076-A1 branch	½" CB203	<i>"plug"</i> — install ½" plug on open connection below valve near PSV F-4.	Piping	HIGH
02007-05	6"-LA-0021-A1-1.5"H	6" CV10	<i>"Doesn't Exist."</i> + cloud — printed 6" valve on lean amine outlet not in field; reconcile. (Eng. Question Q13)	Engineering	HIGH
02007-06	2"-DR-0100-A1 (east)	—	<i>"Removed"</i> + cloud + cut marks — drain header terminus piping removed in field; update drawing / confirm extent.	Engineering	MED
02007-07	2"-AFD-8402-APG-1.5"H-ET	—	<i>"Remove 2x4 & Adjust support"</i> — remove temporary lumber, install permanent support.	Piping	HIGH
02007-08	Spare-filter bridges	PDI F-4B / F-4C	Red tick marks on both PDI balloons — verify gauges installed/functional.	I&E	MED

HAY-PID-02008 Reflux Pumps P-2A/B (400 GPM Amine Plant)Rev E5 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
02008-01	2"-AFD-8406-APG-1.5"H-ET	—	<i>"Add shoes (4)"</i> — install four pipe shoes on antifoam line at skid-edge run.	Piping	HIGH
02008-02	2"-W-8305-A2AS-1.5"H-ET	X-2	<i>"Missing"</i> + pink highlight — install MPLX-supplied 2" static mixer X-2 (Note 1).	Mechanical	HIGH
02008-03	Notes block	—	Pink highlight on note "1) SUPPLIED BY MPLX." — cross-reference to 02008-02 (locate MPLX-supplied material).	Operations	MED
02008-04	2"-RF-0035-A2-1.5"H-ET-SR	LV T-2	Red <i>"¾"</i> + drawn valve below line — install ¾" drain/bleed between block and check.	Piping	HIGH
02008-05	2"-RF-0035-A2-1.5"H-ET-SR	Skid limit	Red <i>"2" / Add"</i> + drawn gate valve — install 2" block valve at Skid #1 limit crossing.	Piping	HIGH

HAY-PID-02009 Amine Still / Reflux Accumulator T-2Rev F12 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
02009-01	12"-AG-0031-A2AS-1.5"H	PSV T-2	<i>"Confirm"</i> — verify pilot-operated PSV set 100 PSIG.	Mechanical	CRITICAL
02009-02	10"-LA-0017-A2-1.5"H	—	Red <i>"1"</i> x3 + drawn valves — install two 1" high-point vents and one 1" drain on lean amine to reboiler.	Piping	HIGH
02009-03	LG T-2A MAG bridle	—	<i>"Missing"</i> — install ½" CB203 vent (top) and ½" CB203 drain (bottom) on gauge bridle.	I&E	HIGH
02009-04	2"-DA-2001-A2-1.5"H-ET, N11	2" CB10	Red <i>"CSC"</i> beside printed CSO — car-seal-closed per field; reconcile CSO vs CSC with engineering. (Eng. Question Q10)	Operations	CRITICAL
02009-05	Nozzle C2 gauges	PI T-2B, TI T-2B	Red range edits <i>"0-100#" / "50-200F"</i> — verify/replace gauges to marked ranges.	I&E	MED
02009-06	2"-DA-2002-A2-1.5"H-ET, N14	2" CB10	Red <i>"CSC"</i> — same car-seal reconciliation at bottom drain.	Operations	CRITICAL
02009-07	T-2 manway M2 (20")	—	<i>"Torque After Inspection"</i> — torque after internal inspection.	Mechanical	HIGH
02009-08	T-2 skirt	—	<i>"Grout"</i> — grout base ring.	Civil	HIGH
02009-09	Lower T-2, N9 area	—	<i>"Install platform"</i> — access platform at reflux line/valve area.	Structural	MED
02009-10	Still-bottoms bridle	LIT T-2B	<i>"Electrical Cover"</i> — install missing transmitter housing cover.	Electrical	HIGH

02009-11	Acid gas PV loop	PY T-2 / PV T-2	<i>"Tubing"</i> — install instrument air tubing (I/P to valve).	I&E	HIGH
02009-12	14"-AG-0033-A2AS-SR PCV station	—	<i>"Install platform"</i> — access platform at 8" acid gas control station.	Structural	MED

HAY-PID-02010 Amine Solution Cooler / Reflux Condenser (AC-1, AC-2)

Rev E

5 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
02010-01	8"-LA-0022-A2 relief riser	PSV AC-1	<i>"Missing"</i> + pink highlight — install PSV AC-1 (2"x1½", set 150 PSIG, check-valve-failure case) and relief route via 3"-V-0102-A2.	Mechanical	CRITICAL
02010-02	PSV AC-1 inlet riser	2" CV10	Red <i>"CSO"</i> — car-seal-open the isolation valve.	Operations	CRITICAL
02010-03	AC-1 louver run	TY AC-1C	<i>"Tubing"</i> — install I/P-to-louver-actuator tubing.	I&E	HIGH
02010-04	AC-2 louver run	TY AC-2C	<i>"Tubing"</i> — install tubing to AC-2 louver actuator.	I&E	HIGH
02010-05	AC-2A/B area	—	<i>"Cleanup Area"</i> — housekeeping.	Operations	MED

HAY-PID-02011 HP Amine Pumps P-3A/B (400 GPM Amine Plant)

Rev G

17 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
02011-01	3"-V-0073-A2	—	Red <i>"3" / Add"</i> + drawn gate valve — install 3" block valve before the drain-system flag.	Piping	HIGH
02011-02	3"-V-0073-A2	PSV P-3C path	<i>"Discharge Pressure? Goes to 165# MAWP Vessel"</i> — engineering must verify 1440 PSIG relief routing to low-pressure drain vessel. (Eng. Question Q3)	Engineering	CRITICAL
02011-03	Relief	PSV P-3C	<i>"Confirm"</i> — verify pilot-operated PSV set 1440 PSIG (blocked flow).	Mechanical	CRITICAL
02011-04	PSV P-3C station	—	<i>"Missing"</i> + pink highlight — install ½"x¾" SS tubing run to P-3A discharge valve string.	I&E	HIGH
02011-05	6"-LA-0025-G2	PIT P-3A2	Red bubble + two drawn root valves at new tap between CV90 and check — relocate PIT P-3A2 tap here; install two small-bore root valves.	I&E	HIGH
02011-06	P-3A discharge cluster	PI/PIT P-3A2	<i>"Wire"</i> + cloud — pull/terminate wiring.	Electrical	HIGH
02011-07	P-3A suction	PI/PIT P-3A1	<i>"Wire"</i> — wire loop.	Electrical	HIGH
02011-08	6"-LA-0024-G2	PI/PIT P-3D	<i>"Wire"</i> + cloud + tap arcs — wire P-3B discharge instruments; relocate tap (mirror of 02011-05).	I&E	HIGH
02011-09	P-3B suction	PI/PIT P-3C	<i>"Wire"</i> — wire loop.	Electrical	HIGH
02011-10	P-3A flanges	—	<i>"Bolt up After Alignment"</i> — final bolt-up of 4" 900# discharge spool and 6" 150# suction flange after alignment.	Mechanical	HIGH
02011-11	P-3B flanges	—	<i>"Bolt up After Alignment"</i> — same at P-3B.	Mechanical	HIGH
02011-12	6"-LA-0025-G2	6" CV90	<i>"Valve Handle"</i> — install missing handwheel at P-3A discharge block.	Piping	HIGH
02011-13	6"-LA-0024-G2	6" CV90	<i>"Valve Handle"</i> — install missing handwheel at P-3B discharge block.	Piping	HIGH
02011-14	6"-LA-0025-F2	FIT/FIC P-3	<i>"Missing"</i> + pink highlight — install FIT P-3 dP transmitter, both impulse manifold valve sets, and tubing on common discharge meter.	I&E	CRITICAL
02011-15	6"-LA-0025-F2	FE P-3	<i>"Confirm"</i> — verify orifice bore 3.5" per datasheet.	I&E	HIGH
02011-16	P-3A suction drop	6" cone strainer	Red doodle at temporary strainer — verify installed; remove after flushing per commissioning plan.	Piping	HIGH
02011-17	P-3B suction drop	6" cone strainer	Same verification at P-3B.	Piping	HIGH

HAY-PID-02012 Reboiler Skid H-1 / E-2 (400 GPM Amine Plant)

Rev F

12 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
02012-01	E-2 level bridles	LG/LT E-2A, E-2B	Large cloud: <i>"Change all of Them to Gate Valves"</i> — replace all bridle isolation valves (½", ¾" CB203; 2" CB10 CSO) with gate valves; count in field before ordering.	Piping	HIGH
02012-02	E-2 platform	—	<i>"Platform Grating"</i> — install missing grating at reboiler platform.	Structural	HIGH

02012-03	E-2 shell relief	PSV E-2	<i>"Missing"</i> + cloud — install 2"x1" thermal PSV, set 150 PSIG.	Mechanical	CRITICAL
02012-04	E-2 bridles	LT E-2A / E-2B	<i>"Wire"</i> — wire both level transmitters.	Electrical	HIGH
02012-05	1" -V-0094-A2 area	TIT E-2E	<i>"Wire"</i> — wire temperature transmitter.	Electrical	HIGH
02012-06	1" -V-0094-A2	—	<i>"Secure"</i> — support/secure 1" drain line.	Piping	HIGH
02012-07	24" -LA-0029-A2-1.5"H	—	<i>"1"ø — Needs plug"</i> — plug open 1" stub next to TE/TW E-2E.	Piping	CRITICAL
02012-08	E-2 supports	—	<i>"Grout supports"</i> — grout at containment.	Civil	HIGH
02012-09	TCV E-2 station	—	<i>"Shim pipe shoes (3)"</i> — shim three shoes at hot-oil control station.	Piping	HIGH
02012-10	E-2 shell (S4-S1)	—	Red-drawn 2" valve at shell connection — install (field addition).	Piping	HIGH
02012-11	TCV E-2 station	—	Red <i>"6"</i> at four station valves — verify actual valve sizes vs drawing; revise drawing or valves. (Eng. Question Q9)	Engineering	MED
02012-12	3" -GH-7008-A2 meter	FQI H-1	<i>"Wire"</i> — wire fuel-gas metering cluster.	Electrical	HIGH

HAY-PID-02013
T.O. for 400 GPM Amine Train & Dehy Train I (VCU-101)

Rev F

14 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
02013-01	2" -GH-7004-A2	—	Red-drawn 2" valve u/s of PCV-100 — install 2" fuel gas isolation valve.	Piping	HIGH
02013-02	2" -FG-2220-A2	FE/FIT-100	<i>"Missing / Need RO + FIT / wire pulled"</i> + pink highlight — install restriction orifice FE-100 and transmitter FIT-100; terminate pulled wire.	I&E	CRITICAL
02013-03	VCU-101 stack	Bird screen	<i>"Remove plastic/wood"</i> — remove shipping covers from stack outlet.	Operations	CRITICAL
02013-04	VCU-101 burner	BS-101/102/103, FR-101/103	<i>"Wire (By Zeeco)"</i> — flame scanner/rod wiring to JB-101, vendor scope.	Vendor	CRITICAL
02013-05	14" -AG-0033-A2AS-SR	—	Red-drawn high-point vent (valve + plug) before KOD-301 — install (likely 1"; verify size).	Piping	HIGH
02013-06	KOD-301	PSV-301, PG-301, LSHH/LSH/LSL/LSLL-301	<i>"Missing wire pulled"</i> + pink highlight — install PSV-301 (set 100 psig, vent SIZE HOLD), PG-301 + ½" MNV, full level-switch stack; terminate wiring.	I&E	CRITICAL
02013-07	KOD-301 top	—	Red-drawn valve at PSV/inlet nozzle connection — install small-bore valve.	Piping	HIGH
02013-08	KOD-301 drain	—	Red-drawn 2" valve at drain termination d/s of flame arrestor — install.	Piping	HIGH
02013-09	14" vent header, NOTE 7	—	<i>"14" Add"</i> + drawn valve + pink highlight — install 14" 150# valve d/s of XV-301.	Piping	CRITICAL
02013-10	KOD-401	PSV-401, PG-401, LSHH...LSLL-401	<i>"Missing wire pulled"</i> + pink highlight — install PSV-401 (set 15 psig), PG-401, level-switch stack; terminate wiring.	I&E	CRITICAL
02013-11	KOD-401 overhead 3" line	—	<i>"Add / 3"</i> + drawn valve d/s of XV-401 — install 3" 150# valve.	Piping	CRITICAL
02013-12	KOD-401 skid	—	<i>"Grout (2)"</i> — grout two base points.	Civil	HIGH
02013-13	VCU-101 / BL-101 base	—	<i>"Grout"</i> — grout base.	Civil	HIGH
02013-14	Blower	BL-101	<i>"wire + ground"</i> — power wiring and grounding.	Electrical	HIGH

HAY-PID-02014
BTEX for 55 MMSCFD Dehy Unit-I (V-7000A, HZZ-1100A, TK-1200A)

Rev F

14 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
02014-01	Fuel gas supply (L)	—	<i>"Tubing"</i> — install ¼" fuel-gas supply tubing run.	I&E	HIGH
02014-02	Diffuser feed	DIF-001A	<i>"Missing Tubing"</i> + pink highlight — install tubing PY-1001A → diffuser at condenser inlet.	I&E	HIGH

02014-03	D — system by-pass vent	—	Cloud: <i>“Missing / Not on ISOs / SKID LIMITS”</i> — fabricate/install 2" NPT bypass-vent piping; add to ISOs. (Eng. Question Q6)	Piping	HIGH
02014-04	2"-PV-2053-A2	—	<i>“Add 2" Valve (CSO)”</i> — install at condenser vapor outlet near PSV-1100A.	Piping	HIGH
02014-05	½" tubing from 2"-GH-7006-A2	PCV-2002A	Red X + pink highlight — install PCV-2002A regulator (set 100 psig) with tubing (see -11).	I&E	HIGH
02014-06	1"-GI-7001-F2AS	—	<i>“1" Add valve”</i> — install 1" valve in flame-arrestor branch.	Piping	HIGH
02014-07	3"-PV-3001-A2	—	<i>“3" Add valve”</i> — install 3" valve at skid edge.	Piping	HIGH
02014-08	TK-1200A	LG-1200A	<i>“Missing”</i> + pink highlight — install gauge glass + 2×¾" isolation, ½" vent, ½" drain valves.	I&E	HIGH
02014-09	TK-1200A east head	—	<i>“plug”</i> — plug open 1½" connection.	Piping	CRITICAL
02014-10	2" NPT bypass-vent run	—	<i>“Missing”</i> + cloud/pink — fabricate/install segment from D vent past HZZ-1100A to BTEX outlet drop near SP-1100A.	Piping	HIGH
02014-11	Motive gas (P) circuit	—	<i>“Missing Tubing”</i> — install ½" tubing riser + header to ABV-1000A and ½"-PV-2054-A2.	I&E	HIGH
02014-12	Condenser liquid outlet	ABV-1200A	<i>“Wire”</i> — wire actuated 3-way valve.	Electrical	HIGH
02014-13	V-7000A vent/condensate	LS-1000A / CV-001	<i>“Tubing”</i> — install tubing for reset/vent circuit to B outlet.	I&E	HIGH
02014-14	2"-DA-2051-A2-1.5"H-ET	—	<i>“2" Add Valve”</i> — install 2" valve in drain to slop tank.	Piping	HIGH

HAY-PID-02115 T.O. for 640 GPM Amine Train & Dehy Train II (VCU-102) Rev F18 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
02115-01	2"-GH-7005-A2	—	Red-drawn 2" valve u/s of PCV-103 — install 2" fuel gas isolation valve.	Piping	HIGH
02115-02	2"-FG-2222-A2	FE/FIT-101	<i>“Missing / RO? / wire pulled”</i> + pink highlight — install FE-101 orifice + FIT-101; terminate pulled wire.	I&E	CRITICAL
02115-03	VCU-102 stack	Bird screen	<i>“Remove Wood/Plastic”</i> — remove shipping covers.	Operations	CRITICAL
02115-04	VCU-102 burner	BS-104/105/106, FR-106	<i>“wire (By Zeeco)”</i> — flame scanner wiring to JB-101, vendor scope.	Vendor	CRITICAL
02115-05	14"-AG-0040-A2AS-SR	—	Red-drawn high-point vent (valve + plug) before KOD-302 — install (verify size).	Piping	HIGH
02115-06	KOD-302	PSV-302, PG-302, LSHH...LSLL-302	<i>“Missing”</i> + pink highlight — install PSV-302 (set 100 psig, vent SIZE HOLD), PG-302 + ½" MNV, level-switch stack.	I&E	CRITICAL
02115-07	KOD-302 14" inlet drop	—	Red-drawn valve at inlet connection — install (size per connection; verify).	Piping	HIGH
02115-08	KOD-302 bridle	LG-302	<i>“Missing”</i> — install gauge glass + 2×1" BAV isolation valves + ½" vents.	I&E	HIGH
02115-09	KOD-302 2" drain	—	Red-drawn valve + termination at open end — install 2" valve (+ flame arrestor per KOD-301 pattern).	Piping	HIGH
02115-10	14"-VF-0023-A2	—	<i>“Install U-bolt”</i> — install 14" U-bolt on vent header.	Piping	HIGH
02115-11	14"-VF-0023-A2, NOTE 7	—	<i>“14" Add”</i> + drawn valve d/s of XV-302 — install 14" 150# valve.	Piping	CRITICAL
02115-12	KOD-402	PSV-402, LSHH...LSLL-402	<i>“Missing”</i> + pink highlight — install PSV-402 (set 15 psig) + level-switch stack.	I&E	CRITICAL
02115-13	KOD-402 overhead → 3"-VF-0003-A2	—	<i>“3" Add”</i> + drawn valve d/s of XV-402 — install 3" 150# valve.	Piping	CRITICAL
02115-14	Detonation arrestors	TE-402 / TE-403	<i>“wire (pulled)”</i> — terminate TE-402/403 (TAHH @ 250°F).	I&E	CRITICAL
02115-15	VCU-102 inlet area	—	<i>“Install 1" IA Plug This Area”</i> — locate open IA connection, install 1" plug.	Piping	HIGH
02115-16	KOD-402 skid	—	<i>“Grout (2)”</i> — grout two base points.	Civil	HIGH
02115-17	VCU-102 / BL-102 base	—	<i>“Grout”</i> — grout base.	Civil	HIGH

02115-18	Blower	BL-102	<i>"Wire + Ground"</i> — power wiring and grounding.	Electrical	HIGH
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HAY-PID-02015A Exhaust Stack / Hot Oil Heater H-27001A (Hot Oil Unit-I)

Rev E

12 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
02015A-01	8"-HM-108A-C2-1.5"H	TE/TI/TY-27001A	<i>"Wire"</i> — wire inlet temperature loop.	Electrical	HIGH
02015A-02	Heater outlet	TE-27002A	<i>"Wire"</i> — wire control thermocouple (TIC-27002A loop).	Electrical	CRITICAL
02015A-03	Heater outlet detail	PI-27002A	Cloud + pink: <i>"Missing / Not on ISO's"</i> — outlet pressure-gauge assembly (0-160 psig, ½" NPT valve, siphon) missing; shipped-loose per Note 8, MPLX installs.	I&E	HIGH
02015A-04	8"-HM-187-C2-1.5"H	—	<i>"Missing PI, Requires Tap on 8"</i> — hot-tap/weld ½" threadolet on 8" outlet line for the PI. (with -03)	Piping	HIGH
02015A-05	Coil feed runs	FE-27003A/ -04A/ -05A	<i>"Torque (By Sigma Thermal ?)"</i> — torque three 4" 300# orifice-flange sets; confirm responsibility. (Eng. Question Q8)	Vendor	CRITICAL
02015A-06	Heater/stack skid	—	<i>"Grout + ground"</i> — grout base, install grounding.	Civil / Elec	HIGH
02015A-07	Combustion blower	—	<i>"Shim Blow Motor & Remove straps"</i> — shim motor, remove shipping straps.	Mechanical	HIGH
02015A-08	Heater electrical	—	<i>"Missing 2" seal off cover"</i> — install.	Electrical	HIGH
02015A-09	Heater instruments	—	<i>"Remove instrumentation plastic"</i> — remove shipping wrap.	Operations	MED
02015A-10	Heater electrical	—	<i>"Missing 2-1" & 2-2" electrical Covers"</i> — install two 1" and two 2" covers.	Electrical	HIGH
02015A-11	Blower 8" flange	—	<i>"Bolt up Blower 8" (by Sigma Thermal ? Need Garlock gasket)"</i> — bolt up; procure 8" Garlock gasket; confirm responsibility. (Eng. Question Q8)	Vendor	CRITICAL
02015A-12	Notes block	Note 8	Pink highlight on "SHIPPED LOOSE BY SIGMA THERMAL AND INSTALLED IN PIPING BY MPLX" — locate all shipped-loose items and install.	Operations	HIGH

HAY-PID-02015B Side Stream Filter & Circulation Pumps (Hot Oil Unit-I)

Rev E

5 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
02015B-01	6"-40-TF-106	—	Red tick at tee — marks tap point for 1" side-stream supply (see -02).	Piping	MED
02015B-02	6"-40-TF-106 → filter	SSI-30001A	Red <i>"1"</i> + drawn field routing from 6" return header to filter — install 1" side-stream supply piping.	Piping	HIGH
02015B-03	Filter hookup	SSI-30001A	Red-drawn loop through filter inlet/outlet (1" 3000# SW nozzles) — re-route/complete filter piping per sketch.	Piping	HIGH
02015B-04	SSI-300017 riser	—	Red trace at header crossing — crossing only (no connection); as-built clarification.	Engineering	LOW
02015B-05	6"-40-TF-110	—	Red <i>"1"</i> + drawn drop — tie side-stream return into discharge header as 1" drop (branch fitting required).	Piping	HIGH

HAY-PID-02015C Expansion Tank TK-45001A (Hot Oil Unit-I)

Rev E

8 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
02015C-01	¾"-GH-2001-A2	PCV-2001	<i>"Missing (¾" valve installed on Adjacent 3" FG pipe)"</i> + pink highlight — blanket-gas regulator station (set 30 psig) not installed as drawn; field deviation exists. Install per design / reconcile. (Eng. Question Q5)	I&E	CRITICAL
02015C-02	To nozzle C3	—	<i>"¾" Tubing Missing"</i> + pink highlight — install blanket-gas tubing across scope breaks to tank top.	I&E	CRITICAL
02015C-03	Nozzles C1, C3	—	Red checks on both 150# RF flanges — complete/verify hookups (with -02).	Piping	HIGH
02015C-04	1½"-PV-0003-A2	PSV-45001A	<i>"Confirm"</i> — verify set 25 PSIG.	Mechanical	CRITICAL
02015C-05	Nozzle C6	LSL-45001A	<i>"Wire"</i> — wire low-level interlock (wiring by MPLX, Note 6).	Electrical	CRITICAL
02015C-06	Nozzle C7	TP-E6	Cloud: <i>"Where to?" / "Don't Exist - Not on ISO's"</i> — test point connection absent and unrouted; engineering to clarify. (Eng. Question Q4)	Engineering	HIGH

02015C-07	8"-HM-187-C2-1.5"H	—	Cloud + <i>"Missing"</i> — install 8" loop/seal (gooseneck) assembly with plug/blind at skid boundary.	Piping	HIGH
02015C-08	8"-HM-180 / 8"-HM-187	FCV-45001A	Cloud + pink: <i>"Doesn't Exist – Not on ISO's" / "Missing"</i> — 8" control-valve station absent entirely; engineering to confirm requirement, then procure/install. (Eng. Question Q4)	Engineering	CRITICAL

HAY-PID-02015D Combustion Air Blower / Burner BRN-20001A (Hot Oil Unit-I) Rev E

3 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
02015D-01	½"-GH-2001-A2 branch	—	<i>"Missing (see 2015C)"</i> + pink highlight — install fuel-gas blanket branch off 3"-GH-7008-A2 (pairs with 02015C-01/-02).	Piping	CRITICAL
02015D-02	1½"-80-NG-204	XV-20003A	Pink highlight — burner main-gas vent valve assembly (1½" XV, FO orifice, vent gooseneck) missing; install (vent piping by MPLX, Note 2).	I&E	CRITICAL
02015D-03	¾"-80-NG-205	XV-20006A	<i>"Missing"</i> (two-headed arrow, also to -02) — pilot-gas vent valve assembly (¾" XV w/ solenoid, FO orifice, gooseneck) missing; install.	I&E	CRITICAL

HAY-PID-02016 Catch Tank TK-2001 / TK-2002 (400 GPM Amine Plant) Rev D

13 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
02016-01	½"-C-7001-F2AS	—	<i>"upsize to 1" and/or support"</i> — upsize condensate line to 1" and/or add supports; engineering to confirm. (Eng. Question Q11)	Engineering	HIGH
02016-02	½"-C-7001-F2AS	—	<i>"Add"</i> + drawn valve before drop to N4 — install isolation valve.	Piping	HIGH
02016-03	Drop to N4 (½")	—	Red fitting stack <i>"1x½"</i> — install union/flange pair, valve, 1"x½" swage at N4.	Piping	HIGH
02016-04	TK-2001 N3	PI-2001	<i>"Install"</i> — install pressure indicator.	I&E	HIGH
02016-05	3" drop to N1	—	<i>"3" Add"</i> + drawn valve — install 3" valve on vent header drop.	Piping	HIGH
02016-06	TK-2001 right head	LG bridle	<i>"Add"</i> — install external level-gauge bridle: standpipe, gauge, ½" top + ½" bottom isolation valves.	I&E	HIGH
02016-07	TK-2001	TI/TW-2001	<i>"Delete"</i> + hatched cloud — remove from scope (with ¾" nozzle).	Engineering	MED
02016-08	TK-2001	LG-2001	Hatched cloud — delete (replaced by external bridle -06).	Engineering	MED
02016-09	TK-2001 left head	—	Hatched cloud over small capped fitting — delete.	Engineering	MED
02016-10	TK-2001 truck-out	—	Red-drawn 2" line: <i>"Elevate, Keep within Containment, Add Truck Connection"</i> — install elevated 2" truck-out w/ 2" valve + capped truck connection.	Piping	HIGH
02016-11	TK-2002 right head	LG-2002	<i>"Missing"</i> + pink highlight — install level gauge (½"x½").	I&E	HIGH
02016-12	TK-2002 drain	—	Red-drawn pump-out w/ <i>"½"x2"</i> swage: <i>"Elevate & Keep within Containment"</i> — install per sketch (2" pipe, valve, cap).	Piping	HIGH
02016-13	TK-2002 header drops	—	Red routing clarifications (break squiggles at 1½"-PV-0002/0003-A2 × 1"-DA-0001-A2 / 1"-V-0094-A2 crossings) — verify as-built tie-in locations; update drawing.	Engineering	MED

HAY-PID-03001 Gas Absorber MAF-3200A (55 MMSCFD Dehy Train-I) Rev D

15 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
03001-01	Upper gauge glass	LG-3201A	<i>"Add Isolation Valves"</i> — install ¾" isolation valve at top connection.	Piping	HIGH
03001-02	Upper gauge glass	LG-3201A	Same note — install ¾" isolation valve at bottom connection.	Piping	HIGH
03001-03	PSV mount	PSV-3200A	Red <i>"1½"</i> + drawn nozzle at vessel wall — verify PSV connection nozzle size 1½" vs printed 1"x1". (Eng. Question Q12)	Engineering	HIGH
03001-04	Glycol drain (F)	RO-3201A	Printed bore struck; red <i>"0.1505"</i> — orifice bore 0.1505"; procure/verify plate.	I&E	HIGH
03001-05	RO-3201A flanges	—	<i>"Bleed Rings"</i> — install bleed ring(s) with bleed valves at orifice flanges.	Piping	HIGH
03001-06	PSV vent stack	SP-3200A	<i>"Secure"</i> — secure/support atmospheric vent stack (SIZE HOLD noted).	Piping	HIGH
03001-07	Lower gauge glass	LG-3200A / TW-3200A	<i>"Add Isolation Valves"</i> — install ¾" isolation valve at upper tap.	Piping	HIGH
03001-08	Lower gauge glass	LG-3200A	Same note — install ¾" isolation valve at lower tap (nozzle G).	Piping	HIGH
03001-09	MAF-3200A skirt	—	<i>"Grout"</i> — grout base.	Civil	HIGH

03001-10	Nozzles E (1" liq outlet), D (1½"-RG-3001-F2)	—	Red hash ticks at vendor termination points (Note 3: piping beyond termination not provided) — complete downstream tie-ins.	Piping	HIGH
03001-11	Nozzles C (lean glycol), H (1"-GH-3012-F2)	—	Red hash ticks at vendor termination points — complete downstream tie-ins.	Piping	HIGH
03001-12	8"-GH-3004-F2	—	Red-drawn 1" plugged valve assembly (stem, valve, flange bars, plug) — install 1" valved+plugged branch (vent/sample) on dry gas line.	Piping	HIGH
03001-13	Liquid outlet (E)	RO-3202A	Bore struck; red <i>"0.1505"</i> — orifice bore 0.1505"; procure/verify plate.	I&E	HIGH
03001-14	IA to LCV-3200A	—	Red-drawn IA detail: 1" plugged tee valve + <i>"plug"</i> , 1" end valve — install per sketch.	I&E	HIGH
03001-15	IA line	—	<i>"U-bolt"</i> — install U-bolt supports on IA supply run.	Piping	HIGH

HAY-PID-03002
Vertical Scrubber MBF-1240A (55 MMSCFD Dehy Train-I)

Rev E

19 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — <i>FIELD TEXT</i> & ACTION REQUIRED	DISC.	PRI.
03002-01	Vessel relief	PSV-1240A	<i>"Confirm"</i> — verify set 1440 PSIG (E orifice).	Mechanical	CRITICAL
03002-02	8"-GH-3005-F2	—	Red-drawn 1" valve on stub at elbow above nozzle B — install 1" high-point vent.	Piping	HIGH
03002-03	8"-GH-3005-F2 meter run	FE/FIT/UIT-1240A	<i>"Missing / Need RO + FIT / Wire pulled"</i> + pink highlight — install orifice plate + FIT-1240A; terminate pulled wire.	I&E	CRITICAL
03002-04	Meter run	TE-1240A	<i>"Wire"</i> — wire temperature element.	Electrical	HIGH
03002-05	FV-1240A manifold	—	Red marks at two ½" bypass/vent valves — install ½" plugs/caps (or valves); confirm intent in field.	Piping	MED
03002-06	Vessel connection	PI-1240A	<i>"Add valve"</i> — install ½" gauge isolation valve.	Piping	HIGH
03002-07	Manway M1 (18" 600#)	—	<i>"Torque"</i> — torque manway.	Mechanical	HIGH
03002-08	Level bridle	LSH-1240A	<i>"Hookup (Wire or Tubing?)"</i> — complete hookup; engineering to confirm electric vs pneumatic. (Eng. Question Q14)	I&E	HIGH
03002-09	Gauge glass	LG-1240A	<i>"Add Isolation valves"</i> — install ¾" valve, upper connection.	Piping	HIGH
03002-10	Gauge glass	LG-1240A	Same — install ¾" valve, lower connection.	Piping	HIGH
03002-11	Vessel skirt	—	<i>"Do we Need this platform?"</i> — RFI to engineering on service platform. (Eng. Question Q15)	Engineering	MED
03002-12	Base ring	—	<i>"Grout"</i> — grout base.	Civil	HIGH
03002-13	2"-DA-3004-F2	RO-1240A	"XX" bore struck; red <i>"0.7394"</i> — orifice bore 0.7394"; procure/verify plate.	I&E	HIGH
03002-14	2"-DA-3019-F2	RO-1241A	Same — bore 0.7394"; procure/verify plate.	I&E	HIGH
03002-15	1"-V-0054-A2	—	<i>"1" Add valve (CSO)"</i> — install 1" CSO valve on PSV-3001 vent to catch tank.	Piping	CRITICAL
03002-16	Below PSV-3001	—	<i>"Add ½" bleed valve (CSO)"</i> — install at 1" CSO isolation (SIZE HOLD noted).	Piping	HIGH
03002-17	Outlet gas GC	AE/AT/AI-1240A	Red cloud over analyzer package + arc to 16" corner — complete GC package hookup (with -18).	I&E	HIGH
03002-18	16"-GH-3006-F2AS	¾" Note-6 tap	<i>"Missing Tap + Tubing"</i> — install analyzer probe tap (GENIE probe) + sample tubing to GC.	I&E	HIGH
03002-19	16" riser / 2"-DA-3003-F2-1.5"H-ET	—	<i>"U/G"</i> breaks + red trace to PCV-3001 — record underground transition; verify slop-header tie-in.	Engineering	MED

HAY-PID-03003
Glycol Region (55 MMSCFD Dehy Train-I)

Rev F

3 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — <i>FIELD TEXT</i> & ACTION REQUIRED	DISC.	PRI.
03003-01	FA-4000A ¾" conn.	TW/TI-4002	<i>"Missing"</i> + pink highlight — install thermowell + temperature indicator.	I&E	HIGH
03003-02	EBC-4001A stack, nozzle L	—	<i>"plug"</i> — install 1" plug in BTEX vapour glow plug connection.	Piping	HIGH
03003-03	BBC-4000A reboiler	LG-4000	<i>"Add Isolation Valves"</i> — install both 1"x½" gauge isolation valves.	Piping	HIGH

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — <i>FIELD TEXT</i> & ACTION REQUIRED	DISC.	PRI.
03004-01	Flash-gas line → 1" -V-0046-A2	1" H5S	<i>"Missing"</i> + cloud — install 1" H5S check valve.	Piping	HIGH
03004-02	MBD-2000A upper gauge	LG-2001A	<i>"Add Isolation valves"</i> — install both ½" tubing-gauge isolation valves.	Piping	HIGH
03004-03	MBD-2000A lower gauge	LG-2000	<i>"Add Isolation Valves"</i> — install both isolation valves.	Piping	HIGH
03004-04	1"A00-PSV-02	PSV-1200	<i>"Add valve. (CSO)" "1"</i> — install 1" CSO valve on charcoal-filter PSV discharge (set 125 psig).	Piping	CRITICAL
03004-05	1"A00-PSV-04	PSV-1300	<i>"Add Valve (CSO)" "1"</i> — install 1" CSO valve on sock-filter PSV discharge.	Piping	CRITICAL
03004-06	1½" A00-GR-04	1½" W2ZU	<i>"valve in wrong spot"</i> / <i>"Has to move"</i> — relocate valve from MAJ-1200A outlet nozzle to marked position on vertical branch.	Piping	HIGH
03004-07	1"A00-PSV-06	PSV-1400	<i>"Add valve (CSO)" "1"</i> — install 1" CSO valve on PSV-1400 discharge.	Piping	CRITICAL
03004-08	1½"-RG-3001-F2	—	<i>"1½" Add Valve"</i> — install 1½" isolation valve at Cimarron/MPLX scope break (rich glycol).	Piping	HIGH
03004-09	1½" -LG-3001-F2	—	<i>"1½" Add Valve"</i> — install 1½" isolation valve at scope break (lean glycol).	Piping	HIGH

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — <i>FIELD TEXT</i> & ACTION REQUIRED	DISC.	PRI.
04001-01	TK-41001	LIT/LSH-4001	<i>"Wire"</i> — wire level transmitter + switch.	Electrical	HIGH
04001-02	TK-41002	LIT/LSH-4002	<i>"Wire"</i> — wire.	Electrical	HIGH
04001-03	TK-41003	LIT/LSH-4003	<i>"Wire"</i> — wire.	Electrical	HIGH
04001-04	TK-41004	LIT/LSH-4004	<i>"Wire"</i> — wire.	Electrical	HIGH
04001-05	PVRV risers, C3 (×4)	Enardo ES-660-HF	<i>"6X4"</i> ×4 — install 6"x4" transition/reducer at each tank's PVRV riser.	Piping	HIGH
04001-06	Tank bottoms (×4)	—	Red-drawn 1" valve at each tank near manway — install four 1" drain/water-draw valves.	Piping	HIGH
04001-07	C4 nozzles (×4)	—	Red triangle at each 4" side nozzle — verify blinded; install 4" blind flanges if open.	Piping	HIGH
04001-08	2"-DA-2101-A2-1.5"H-ET	—	Red-drawn new 2" HT drain line + 2" valve + capped riser at containment (P&ID ref blank) — install; engineering to assign tie-in/sheet ref.	Piping	HIGH
04001-09	2"-DA-2[6]81-A2-1.5"H-ET [digits unclear]	—	Second red-drawn 2" HT drain line + 2" valve, tie-in "HAY-P&ID 02013" — install; verify line number with engineering.	Piping	HIGH
04001-10	Drain headers	—	Red tie-in routing arrows into 2"-DA-3003-A2 / 4"-DA-3023-A2 / 2"-DA-2051-A2 junctions — verify as-built tie-in points; update drawing.	Engineering	MED
04001-11	Containment, south of TK-41002	—	<i>"Remove wood"</i> — remove dunnage from containment.	Operations	MED
04001-12	4" transfer line	ESD-4001, ZSC/ZSO-4001, SV-4001	<i>"Wire"</i> — wire ESD valve solenoid + position switches.	I&E	CRITICAL
04001-13	All four tanks	—	<i>"Ground All Tanks"</i> — ground/bond TK-41001-41004.	Electrical	CRITICAL
04001-14	Grounding station	—	<i>"Missing"</i> — supply & install grounding station assembly.	Electrical	CRITICAL
04001-15	4"-DA-4010-A2-1"H-ET	—	<i>"Install Enviro Box"</i> — install environmental containment box at 4" valve/spec-blind station.	Piping	MED

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — <i>FIELD TEXT</i> & ACTION REQUIRED	DISC.	PRI.

07001-01	VS-70004 wet receiver	PSV-7005	<i>“Remove”</i> + cloud + X — remove PSV-7005 manifold (set 155 psig) and adjoining PI-7004/½" valve. Engineering to confirm relief philosophy change. (Eng. Question Q16)	Engineering	CRITICAL
07001-02	VS-70004 top, 6" nozzle	—	<i>“VENT”</i> + drawn gooseneck along 1½"-IA-7036-L2 — install atmospheric vent in lieu of removed PSV.	Piping	CRITICAL
07001-03	½"-IA-7040-L2	—	Red X-hatch — delete/remove line + ½" SS tube run below VS-70004.	Piping	HIGH
07001-04	VS-70004 bottom drain	Timer drain	Red <i>“1”</i> + magenta — install/upsized drain train to 1".	Piping	HIGH
07001-05	VS-70001 dry receiver	PSV-7003	<i>“Remove”</i> + cloud + X — remove PSV assembly (set 155 psig). (Eng. Question Q16)	Engineering	CRITICAL
07001-06	VS-70001 top nozzle	—	<i>“Vent”</i> + <i>“←PI”</i> — install gooseneck vent + pressure gauge at top (1½"-IA-7032-L2).	Piping	HIGH
07001-07	VS-70001 shell	PI-7001	Red cloud — relocate PI-7001 to top nozzle (with -06).	I&E	HIGH
07001-08	VS-70001 bottom drain	Timer drain	Red <i>“1”</i> — upsized drain train to 1".	Piping	HIGH
07001-09	VS-70002 dry receiver	PSV-7001	<i>“Remove”</i> + cloud + X — remove PSV assembly. (Eng. Question Q16)	Engineering	CRITICAL
07001-10	VS-70002 top nozzle	—	<i>“Vent”</i> + <i>“PI”</i> — install vent + PI (1½"-IA-7035-L2).	Piping	HIGH
07001-11	VS-70002 shell	PI-7002	Red cloud — relocate to top (with -10).	I&E	HIGH
07001-12	VS-70002 bottom drain	Timer drain	Red <i>“1”</i> — upsized drain train to 1".	Piping	HIGH
07001-13	VS-70001 / VS-70002	Spare nozzles	<i>“Missing plugs”</i> — install NPT plugs in open spare nozzles.	Piping	CRITICAL
07001-14	3"-IA-7006-L2	PSV-7002	<i>“Missing”</i> + large cloud — install COMPLETE relief station: PSV-7002 (1"x2", set 121 psig), 2" CSO block, 2"x1" reducer, ½" bleed, vent pipe.	Mechanical	CRITICAL
07001-15	3"-IA-7006/7007-L2	PCV-7001/-7002	<i>“Missing”</i> — install four missing ½" valves at regulator manifold (set 110 psig).	Piping	HIGH
07001-16	3"-IA-7006-L2	—	<i>“3" U-bolt”</i> — install at air distribution flag.	Piping	HIGH
07001-17	Vessels + building	—	<i>“Ground 3 Vessels + Building”</i> — install grounding.	Electrical	HIGH
07001-18	½" SS tube runs	—	<i>“Missing ½" Tubing”</i> + magenta — install tubing: skid boundary to timer drains, below VS-70004, and to TK-70001 hookup.	I&E	HIGH
07001-19	TK-70001	Hose connection	Red cloud + strikes over <i>“HOSE CONNECTION”</i> /2" valve/ 2"-IA-7042-L2 — remove hose connection; update drawing.	Piping	HIGH
07001-20	TK-70001	LG-7001	<i>“Missing”</i> + magenta — install level gauge + two 1" isolation valves.	I&E	HIGH
07001-21	TK-70001 discharge	—	<i>“Elevate & Keep Within Containment” “2”</i> — install elevated 2" line w/ valve, end hook inside containment.	Piping	HIGH
07001-22	TK-70001 outlet	—	Red <i>“1X2”</i> + drawn expander at containment boundary — install 1"x2" swage.	Piping	HIGH

HAY-PID-07002
JT Fuel Skid (V-100/200/300, E-400A/B, P-500A-D)

Rev F

17 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
07002-01	4"-GH-3007-F2	—	<i>“Add Support @ Double 90°”</i> — install pipe support at vertical double-elbow drop (near PI/PCV-7101/7102).	Piping	HIGH
07002-02	Methanol pumps title block	P-500A–D	Red cloud + <i>“**”</i> — all four methanol pumps flagged; see Q7 (meaning of asterisks to be confirmed).	Engineering	MED
07002-03	Pump discharge	P-500A	Red <i>“*”</i> at discharge assembly — pump flagged for attention (Q7).	Engineering	MED
07002-04	Pump discharge	P-500B	Red <i>“*”</i> — flagged (Q7).	Engineering	MED
07002-05	Pump discharge	P-500C	Red <i>“*”</i> — flagged (Q7).	Engineering	MED
07002-06	Pump discharge	P-500D	Red <i>“*”</i> — flagged (Q7).	Engineering	MED

07002-07	Skid connection E	1" -PL-7002-A2	Cloud: <i>“Connect, IA Currently Hooked up Here”</i> — instrument air is wrongly connected to the methanol inlet port; remove IA, connect methanol supply. Cross-connection hazard.	Piping	CRITICAL
07002-08	Skid conn. E → SG-7	—	<i>“Connect + support OVHD 1" IA”</i> — run and support 1" overhead instrument-air line to the skid (correct IA landing).	Piping	CRITICAL
07002-09	Supply gas to instrumentation	PCV-501, SG-1..SG-6	Red <i>“***”</i> at header — cross-reference flag (with -02/-08); confirm intent.	Engineering	MED
07002-10	V-100 overhead	PCV-100A, PI-101A	Cloud + <i>“Missing”</i> — install instrument-gas regulator set: PCV-100A, PI-101A (0-150#), ¼" valve, tubing, TO SG-7 takeoff.	I&E	HIGH
07002-11	Above LCV-100	—	Red <i>“¼”</i> + drawn valve — install ¼" valve upstream of LCV-100 (size ¼" vs ¾" to be verified).	Piping	HIGH
07002-12	Below LCV-100	—	Red <i>“¼”</i> + drawn valve above 1" DSS spec break — install ¼" bleed.	Piping	HIGH
07002-13	D/s LCV-200	—	Red <i>“¼”</i> + drawn valve at 2" DSS break — install ¼" valve.	Piping	HIGH
07002-14	D/s LCV-300	—	Red <i>“¾”</i> + drawn valve at 2" DSS break — install ¾" valve (size to be verified).	Piping	HIGH
07002-15	V-300 overhead 2" -PG-300-03	Methanol inj. pt	Cloud: <i>“not connected inside skid.”</i> — complete ¼" methanol-injection tie-in from pump 4.	Piping	HIGH
07002-16	3" -PG-104-A (E-400A N4)	—	Red highlighter trace, no text — segment flagged; verify installation/routing.	Piping	MED
07002-17	3" -PG-104-B (E-400B N4)	—	Red highlighter trace — verify installation/routing.	Piping	MED

HAY-PID-07003 FG Coalescing Filter F-71001

Rev E

13 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
07003-01	PSV tailpipe	PSV-7101	<i>“Missing”</i> + pink highlight — install PSV vent tailpipe TO ATM (SIZE HOLD — engineering to release).	Piping	CRITICAL
07003-02	Relief	PSV-7101	<i>“Confirm”</i> — verify set 275 PSIG (fire, 1D2).	Mechanical	CRITICAL
07003-03	2" PSV inlet (N9/N10)	—	<i>“Add support”</i> — support 2" inlet line.	Piping	HIGH
07003-04	4" -GH-7001-A2	—	<i>“Add Support (2)”</i> — two supports upstream of N1.	Piping	HIGH
07003-05	DP installation	PDI-7101	Cloud: <i>“By Valley.”</i> / <i>“Missing”</i> — install PDI-7101 assembly (manifold, ½" valves, tubing, 2"x½" taps); vendor “Valley” scope.	Vendor	HIGH
07003-06	Gauge standpipe (N5)	—	Red-drawn vent valve at top — install.	Piping	HIGH
07003-07	Stub below N6	—	Red-drawn valve + plug — install drain valve + plug above 2" -LH-7002-A2-1.5"H-ET.	Piping	HIGH
07003-08	Nozzle N8 stub	—	Red valve symbol (~¾", partially legible) at KMTD/MPLX boundary — install valve; verify size.	Piping	HIGH
07003-09	Stub below N7	—	Red-drawn valve + plug next to RO-7102 — install drain valve + plug.	Piping	HIGH
07003-10	2" -LH-7002-A2-1.5"H-ET	RO-7101	“XX” struck; red <i>“0.1031”</i> — orifice bore 0.1031"; procure/verify plate.	I&E	HIGH
07003-11	N4 outlet	RO-7102	Same — bore 0.1031"; procure/verify plate.	I&E	HIGH
07003-12	Meter run	FE/FIT/MVT-7101, AE/AT-7101	<i>“wire”</i> — wire meter-run instrumentation (heated enclosure).	Electrical	HIGH
07003-13	F-71001 base	—	<i>“Install Platform”</i> — access platform.	Structural	MED

HAY-PID-07004 Blowcase System V-71001...V-71004

Rev E

16 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
07004-01	V-71001 bottom outlet	1" valve	<i>“Remove”</i> + X — remove 1" valve (into 2" drain manifold).	Piping	HIGH
07004-02	V-71002 bottom outlet	1" valve	<i>“Remove”</i> + X — remove 1" valve (to 2" -DR-7002-F2AS).	Piping	HIGH

07004-03	1"-G-0002-F2AS motive gas	—	"1" Add" + drawn valve at battery limit — install 1" valve.	Piping	HIGH
07004-04	IA at V-71002 regulators	PCV-71007/71008	"Add 1" Valve" — install 1" IA isolation valve.	Piping	HIGH
07004-05	1" PRO GAS line	—	"1" Add" + drawn valve at PRO GAS MPLX boundary — install 1" valve.	Piping	HIGH
07004-06	2" drain from V-102 (HAY-PID-02105)	—	"Add 2"" + drawn valve at A2AS F2AS spec break — install 2" valve.	Piping	HIGH
07004-07	2"-DR-7002-F2AS	—	"2" / Add" + drawn valve + red routing to drain header — install 2" valve; complete routing.	Piping	HIGH
07004-08	2"-DR-7001-F2AS	2" check valve	"Installed Backwards" — check valve at F2 F2AS break is reversed; reinstall in correct flow direction.	Piping	CRITICAL
07004-09	2" drain nr. V-71003 inlet	—	"Add support" — install 2" pipe support at MPLX PRO GAS boundary.	Piping	HIGH
07004-10	V-71003 sensing line	PC/PV-71003C	Red-drawn valve + "¼"" — install ¼" valve in sensing line.	I&E	HIGH
07004-11	LG-71003 gauge drain	—	Red tick + triangle at 1"x¾" tap — install ¾" bull plug / verify drain valve.	Piping	HIGH
07004-12	LG-71004 gauge drain	—	Red marks below lower gauge valve — install ¾" drain valve + plug.	Piping	HIGH
07004-13	V-71003/71004 outfall	Open drain	Cloud + "← Connect Here" — tie open drain into header at marked point (near 2"-DR-7004-F2AS).	Piping	HIGH
07004-14	IA regulator station	PCV-71007/71008	"support @ P.R." — install support at pressure-reducing station.	Piping	HIGH
07004-15	ESD valve group	SV/ESD-71004, ZS0/ZSC-71004	"Secure 1" IA pipe + Wire + Tubing" — secure IA pipe; complete wiring & tubing at ESD group.	I&E	CRITICAL
07004-16	2"-DR-7003-F2AS	—	"U/G" + arc upstream of outlet-header flag — record underground routing (mate of 01001-38).	Engineering	LOW

HAY-PID-07005
Methanol Tank TK-71001

Rev C

3 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
07005-01	TK-71001 bottom	—	"Ground" — install tank grounding/bonding (flammable service).	Electrical	CRITICAL
07005-02	1"-PL-7001-A2	—	"support" — add pipe support(s) at mid-span of long 1" methanol run.	Piping	HIGH
07005-03	1"-PL-7001-A2 flag	—	Cloud: "Connect to JT Skid" — tie methanol line in at JT skid (companion of 07002-07).	Piping	CRITICAL

HAY-PID-07006
Generator Fuel Gas (Generators I / II / III)

Rev D

8 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — FIELD TEXT & ACTION REQUIRED	DISC.	PRI.
07006-01	3"-GH-7010-A2 , d/s PCV-7003	PSV-7006	"Missing" + pink highlight — install PSV-7006 (PCV-failure case, SIZE HOLD) + vent to ATM.	Mechanical	CRITICAL
07006-02	3"-GH-7602-A2	Gen I branch	"U/G" — record underground takeoff (Note 3).	Engineering	LOW
07006-03	3"-GH-7602-A2	Gen I strainer	"Add Valve" — install 3" valve at Y-strainer assembly.	Piping	HIGH
07006-04	3"-GH-7601-A2	Main header	"U/G" — record underground segment.	Engineering	LOW
07006-05	3"-GH-7603-A2	Gen II branch	"U/G" — record underground takeoff.	Engineering	LOW
07006-06	3"-GH-7603-A2	Gen II strainer	"Add valve" — install 3" valve at Y-strainer.	Piping	HIGH
07006-07	3"-GH-7604-A2	Gen III branch	"U/G" — record underground takeoff.	Engineering	LOW
07006-08	3"-GH-7604-A2	Gen III strainer	"Add Valve" — install 3" valve at Y-strainer.	Piping	HIGH

HAY-PID-08001

Amine Makeup Tank TK-81003 / P-81002

Rev E

1 item

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — <i>FIELD TEXT</i> & ACTION REQUIRED	DISC.	PRI.
08001-01	2" -LA-8018-A2-1.5"H-ET	P-81002 discharge	<i>"support"</i> — install pipe support on 2" heat-traced discharge line.	Piping	HIGH

HAY-PID-08002

TEG Glycol Makeup Tank TK-81005 / P-81005

Rev E

1 item

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — <i>FIELD TEXT</i> & ACTION REQUIRED	DISC.	PRI.
08002-01	2" -LG-3003-A2	P-81005 discharge	<i>"support"</i> — install 2" pipe support.	Piping	HIGH

HAY-PID-08003

Chemetherm Makeup Tank TK-81007 / P-81007

Rev E

1 item

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — <i>FIELD TEXT</i> & ACTION REQUIRED	DISC.	PRI.
08003-01	2" -O-8207-A2	P-81007 discharge	<i>"support"</i> — install 2" pipe support.	Piping	HIGH

HAY-PID-08004

Water Makeup Tank TK-81009 / TK-81010

Rev E

5 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — <i>FIELD TEXT</i> & ACTION REQUIRED	DISC.	PRI.
08004-01	2" -W-8303-A2AS	TK-81010 fill line	<i>"Delete"</i> + large hatched cloud — remove/do-not-install entire truck-unload fill line incl. 2" check, 2" ball valve, truck connection, 2"x3" roof fill.	Piping	HIGH
08004-02	TK-81010 shell	LG	<i>"Missing"</i> + red-drawn gauge column — install external level gauge with 1" top & bottom isolation valves.	I&E	HIGH
08004-03	TK-81010 bottom outlet	3" assembly	<i>"Delete"</i> + cloud — remove existing 3" bottom outlet valve assembly.	Piping	HIGH
08004-04	TK-81010 new manifold	—	Red-drawn replacement: 2" header off bottom outlet w/ 2" break, 2" valve, hose/truck-fill stub, LG tie-in (1" valve), and new 1" suction run to P-81010 — install per sketch; verify fitting types in field.	Piping	HIGH
08004-05	2" -W-8304-A2AS-1.5"H-ET	P-81010 suction	<i>"Delete"</i> + cloud — remove/do-not-install old heat-traced suction line (superseded by -04).	Piping	HIGH

HAY-PID-08005

Anti Foam Tank TK-81011 / P-81011

Rev B

6 items

ITEM	LINE / EQUIPMENT	TAG	DESCRIPTION — <i>FIELD TEXT</i> & ACTION REQUIRED	DISC.	PRI.
08005-01	P-81011 air supply	—	<i>"Tubing"</i> — run air-supply tubing from 2" valve (PCV-8401 station) to pump air inlet.	I&E	HIGH
08005-02	2" -AF0-8402-APG-1.5"H-ET	2" check valve	<i>"Missing"</i> + red oval — install 2" check valve downstream of discharge flex.	Piping	HIGH
08005-03	2" -AF0-8402-APG-1.5"H-ET	—	<i>"support"</i> — install pipe support mid-run (heat-traced).	Piping	HIGH
08005-04	End of discharge run	—	Red-drawn valve + <i>"2"</i> (+ “inst...” partially legible) — install 2" isolation valve upstream of anti-foam distribution split.	Piping	HIGH
08005-05	Distribution split → P-1A/B	2" ball valve	<i>"Missing."</i> + cloud — install 2" ball valve on through-run to HAY-PID-02007.	Piping	HIGH
08005-06	Distribution area	—	<i>"Need to Correct Iso To Match P&ID"</i> — revise piping isometric to match P&ID.	Engineering	LOW

4 Material Takeoff (BOM) — Missing Items Only

Only material a redline shows as missing / to be installed. Where the drawing does not state rating or material, the entry reads **“Verify in Piping Material Specification”** (V-PMS) against the line class shown — no specifications have been invented. Quantities marked ± require field count before ordering.

QTY	DESCRIPTION	SIZE	RATING	MATERIAL	SPEC	TAG	LINE NO.
1	PSV, complete w/ inlet block & vent tailpipe	V-PMS	Set per V-22001 (1435 PSIG design)	V-PMS	A2AS vent	PSV V-22001	3" -GH-00

1	PSV assembly: PSV + 2" CSO block + 2"x1" reducer + ½" bleed + 2" ATM tail	2"x1"	Set 1440 PSIG (fire)	V-PMS	V-PMS	PSV - 1100	S-1100
1	PSV, thermal relief	2"x1"	Set 150 PSIG	V-PMS	V-PMS	PSV E-2	E-2 shel
1	PSV (check-valve-failure case)	2"x1½"	Set 150 PSIG	V-PMS	A2	PSV AC-1	8" - LA-00
2	PSV, KO drum (fire) w/ vent to safe location	SIZE HOLD	Set 100 PSIG	V-PMS	V-PMS	PSV-301 / PSV-302	KOD-30
2	PSV, KO drum (fire) w/ vent to safe location	SIZE HOLD	Set 15 PSIG	V-PMS	V-PMS	PSV-401 / PSV-402	KOD-40
1	PSV relief station complete (PSV + 2" CSO block + 2"x1" reducer + ½" bleed + vent)	1"x2"	Set 121 PSIG	V-PMS	L2	PSV-7002	3" - IA-70
1	PSV (PCV failure) + ATM vent	SIZE HOLD	Set 5 PSIG system	V-PMS	A2	PSV-7006	3" - GH-70
1	PSV vent tailpipe to ATM	2"	V-PMS	V-PMS	A2	PSV T-1	2" - V-003
1	PSV vent tailpipe to ATM	SIZE HOLD	V-PMS	V-PMS	V-PMS	PSV-7101	F-71001
1	Choke nipple (restriction orifice)	1", ⅜" bore	V-PMS	V-PMS	F2AS	R0 V-1	1" - C-000
2	Orifice plate, bore 0.1031"	2" line	V-PMS	V-PMS	A2	R0-7101 / R0-7102	2" - LH-70
2	Orifice plate, bore 0.7394"	2" line	V-PMS	V-PMS	F2	R0-1240A / R0-1241A	2" - DA-30
2	Orifice plate, bore 0.1505" + bleed ring set w/ bleed valves	per line	6000# taps	V-PMS	F2	R0-3201A / R0-3202A	MAF-32
2	Flow element (RO) + FIT transmitter	2" line	V-PMS	V-PMS	A2	FE/FIT-100, FE/FIT-101	2" - FG-22
1	Orifice plate + FIT (meter run exists)	8" line	V-PMS	V-PMS	F2	FE/FIT-1240A	8" - GH-30
1	dP flow transmitter + 2 impulse manifold valve sets + tubing	—	V-PMS	—	F2	FIT P-3	6" - LA-00
2	Cone strainer, temporary	4"	V-PMS	V-PMS	F2AS	—	4" - LH-00
2	Flow device "4" FD"	4" — SIZE HOLD	V-PMS	V-PMS	F2AS	—	same
1	Static mixer (MPLX-supplied — locate)	2"	V-PMS	V-PMS	A2AS	X-2	2" - W-830
1	Control valve station w/ actuator (existence to be confirmed by engineering first)	8"	V-PMS	V-PMS	C2	FCV-45001A	8" - HM-18
1	Blanket-gas regulator + ¼" fittings	¼"	Set 30 PSIG	V-PMS	A2	PCV-2001	¼" - GH-20

±4	Valve, ¼" (LCV stations, sensing lines)	¼"	V-PMS	V-PMS	DSS/per class	—	JT skid,
3	Valve, isolation	1½"	V-PMS	V-PMS	F2	—	1½" -RG-3
±15	Valve (block/gate/ball/check) incl. CSO where marked	2"	V-PMS	V-PMS	per line class	—	various
±6	Valve, block	3"	V-PMS	V-PMS	A2 / V-PMS	—	3" -V-007
1	Check valve	1" (H5S)	V-PMS	V-PMS	A00	—	flash ga
Field count	Gate valves to replace E-2 bridle valves (½", ¾" CB203; 2" CB10 CSO)	½"-2"	V-PMS	V-PMS	CB203/CB10 dims	LG/LT E-2A/B	E-2 brid
2	Valve spec change GVTONT → CB10 (CSO)	2"	V-PMS	V-PMS	CB10	LG V-2A bridle	V-2 N8/I
2	Handwheel/valve handle	6" CV90	—	—	—	P-3A/B discharge	6" -LA-00
1	Blind flange + gasket + stud bolts/nuts	3"	V-PMS	V-PMS	F2AS	V-22001 bridle drain	2" -LH-00
±4	Blind flange (verify open) + gaskets/bolting	4"	V-PMS	V-PMS	—	Slop tank C4 nozzles	TK-4100
4	Reducer/transition, PVRV riser	6"x4"	V-PMS	V-PMS	—	Enardo risers	TK-4100
1	Reducer	3"x2"	V-PMS	V-PMS	F2AS	—	3" -LH-00
1	Reducer + cap (LIT mount)	3"x2"	V-PMS	V-PMS	—	LIT-1004	V-22001
1	Expander/swage	1"x2"	V-PMS	V-PMS	L2	TK-70001 outlet	—
1	Swage + union/flange pair + valve	1"x½"	V-PMS	V-PMS	F2AS	TK-2001 N4 drop	½" -C-700
lot	SS instrument tubing + compression fittings (PDT T-1 column run ~55 ft; degassers; P-3C ½"x¾"; analyzer sample; louver actuators ×3; blanket gas; motive gas circuit; timer drains)	¼"-½"	—	316 SS	—	—	various

lot	Plugs: 1" (3+), 1½" (1), ½" (2+), ¾" bull plugs (2), NPT assortment for spare nozzles	½"-1½"	V-PMS	V-PMS	per class	—	02012-07004-1
2	Truck/hose connection w/ cap (+1 hose/quick-connect stub, TK-81010)	2"	V-PMS	V-PMS	—	TK-2001, TK-2002, TK-81010	—
1	Environmental containment (drip) box	—	—	—	—	—	4" -DA- 40
1	Analyzer probe tap (GENIE) + sample tubing	¾"	V-PMS	316 SS tubing	F2AS	AT -1240A	16" -GH- 3
1	Transmitter electrical housing cover	—	—	—	—	LIT T-2B	—

5 Engineering Questions

1. **Q1 — Slug catcher metering (HAY-PID-01001):** field asks *“Installing Orifice Meter or RO + FIT?”* on 12"-GH-0010-F2AS . Confirm metering design so the meter run can be completed. (01001-34)
2. **Q2 — Coalescing filter inlet (HAY-PID-02001):** same question — *“RO + FIT ?”* sketched on 10"-GH-0028-F2AS . (02001-01)
3. **Q3 — HP pump relief routing (HAY-PID-02011):** *“Discharge Pressure? Goes to 165# MAWP Vessel”* — PSV P-3C (set 1440 PSIG) relieves via 3"-V-0073-A2 to the drain system. Verify overpressure protection of the receiving vessel. **Process-safety critical.** (02011-02)
4. **Q4 — Hot Oil expansion tank (HAY-PID-02015C):** FCV-45001A station and test point TP-E6 *“don’t exist / not on ISO’s”*. Confirm whether FCV-45001A is required (procure if so — 8" control valve, long lead) and where TP-E6 routes. (02015C-06/-08)
5. **Q5 — Blanket gas deviation (HAY-PID-02015C):** PCV-2001 station missing; field notes *“¾" valve installed on Adjacent 3" FG pipe”* instead. Accept deviation or install per design? (02015C-01)
6. **Q6 — BTEX bypass vent (HAY-PID-02014):** 2" NPT system-bypass-vent piping missing and *“Not on ISOs”*, crosses skid limits — issue ISO/routing. (02014-03/-10)
7. **Q7 — JT skid asterisks (HAY-PID-07002):** red * on P-500A-D and ** on the pump title block / supply-gas header — clarify field intent (deadhead check? alignment? vendor issue?). (07002-02...06, -09)
8. **Q8 — Sigma Thermal scope (HAY-PID-02015A):** orifice-flange torquing (*“By Sigma Thermal ?”*) and 8" blower bolt-up responsibility — confirm vendor vs MPLX scope. (02015A-05/-11)
9. **Q9 — TCV E-2 station sizes (HAY-PID-02012):** red “6” marked at four station valves vs printed sizes — verify installed sizes; revise drawing or hardware. (02012-11)
10. **Q10 — Car seals on T-2 drains (HAY-PID-02009):** field marks CSC where drawing shows CSO at N11/N14 drains — confirm car-seal philosophy. (02009-04/-06)
11. **Q11 — Condensate line size (HAY-PID-02016):** *“upsized to 1" and/or support”* for ½"-C-7001-F2AS — confirm resolution. (02016-01)
12. **Q12 — PSV-3200A nozzle (HAY-PID-03001):** red 1½" vs printed 1"x1" NPT — confirm PSV connection size. (03001-03)
13. **Q13 — “Doesn’t exist” valves (HAY-PID-02007 , 02005):** printed 1" C1203F LSD valve and 6" CV10 lean-amine valve absent in field; PSV F-2 tailpipe 1" CSO absent. Install per drawing or revise drawing? (02007-03/-05, 02005-10/-11)
14. **Q14 — LSH-1240A hookup (HAY-PID-03002):** *“Wire or Tubing?”* — confirm electric vs pneumatic. (03002-08)
15. **Q15 — Platform need (HAY-PID-03002):** *“Do we Need this platform?”* at MBF-1240A. (03002-11)
16. **Q16 — Air-system relief philosophy (HAY-PID-07001):** field removes PSV-7001/-7003/-7005 and adds open vents + PIs at receiver tops, and installs missing PSV-7002 on the header. Confirm the relief redesign is engineered & code-compliant before executing removals. **Do not remove reliefs without engineering sign-off.** (07001-01/-05/-09, -14)
17. **Q17 — SIZE HOLD releases:** engineering to release held sizes blocking procurement: PSV-301/-302/-401/-402 vents, PSV-7006, PSV-7101 tailpipe, SP-3200A vent stack, 4" FD devices (01001-25/-26), 2" FC (01003-20), 3x2 LT reducer verification.
18. **Q18 — New slop-tank drain lines (HAY-PID-04001):** two red-drawn 2" HT lines; one tie-in P&ID ref left blank, second line number partially legible (“2"-DA-2[6]81-A2”) — assign line numbers/sheet refs and confirm tie-in points. (04001-08/-09/-10)

6 Procurement List

Condensed order list from the BOM. **LL** = likely long-lead; **HOLD** = blocked pending engineering answer (Section 5).

PRIORITY	ORDER	QTY	NOTES
CRITICAL	PSVs: V-22001, PSV-1100, PSV E-2, PSV AC-1, PSV-301/-302/-401/-402, PSV-7002 station, PSV-7006	10	LL — sets/orifices per data sheets; 4 vents HOLD (Q17)
CRITICAL	14" 150# valves (VF headers); 3" 150# valves (VF)	2 + 2	LL
CRITICAL	KO-drum level-switch stacks; PG-301/302 w/ MNV	4 stacks + 2	—
CRITICAL	Burner vent assemblies XV-20003A / XV-20006A; PCV-2001; blanket ¼" tubing	2 + 1 + lot	Sigma Thermal parts — confirm scope (Q8)

CRITICAL	FCV-45001A 8" control valve station	1	LL + HOLD (Q4)
HIGH	Orifice plates: 0.1031" x2, 0.7394" x2, 0.1505" x2 (+bleed rings); RO V-1 choke nipple 3⁄8" bore; FE/FIT-100/-101/-1240A; FIT P-3 + manifolds	—	bores are field-specified — verify against datasheets
HIGH	Valve package: 1⁄4" x4, 1⁄2" x10±, 3⁄4" x12±, 1" x16±, 1½" x3, 2" x15±, 3" x6±, gate-valve swap for E-2 bridles (field count), 2" CB10 x2, 1" H5S check, 2" check (anti-foam)	~70	order to line-class specs — Verify in PMS
HIGH	Level gauges x6; PI/PG x5; TI/TW x2; PDI-7101 assembly; PCV-2002A; PCV-100A set	—	—
HIGH	4" cone strainers x2 + 4" FD x2	4	FD HOLD (Q17)
HIGH	SS tubing 1⁄4"-1⁄2" + fittings (bulk); instrument cable (bulk)	lot	—
HIGH	Supports: shoes x14±, U-bolts x8±, misc supports/braces x12±, shim stock	lot	—
HIGH	Pipe & fittings for field-routed lines (1"-2", incl. 1.5"H-ET heat trace on two slop drains); 8" loop/seal; 1⁄2" threadolet (8" C2); reducers 6"x4" x4, 3"x2" x2, 1"x2", 1"x1⁄2"; blind flanges 3" x1, 4" x4; gaskets & stud bolts	lot	heat trace + insulation to match -ET lines
HIGH	Grounding: station assembly, cable, lugs, rods; electrical covers (1" x2, 2" x2), 2" seal-off cover; 8" Garlock gasket	lot	—
MED	Platform steel + grating (7± locations); enviro box; truck connections x2 + caps; plugs assortment	lot	one platform HOLD (Q15)

7 Missing Information / Cannot Be Verified From Drawings

- **SIZE HOLD items** (drawing-stated): PSV-301/-302/-401/-402 vents, PSV-7101 tailpipe, PSV-7006, SP-3200A vent stack, 4" FD (01001-25/-26), 2" FC (01003-20) — sizes must come from engineering.
- **Unmarked sizes on red-drawn valves:** 01001-13/-16/-17, 02006-02, 02013-07, 02115-07, 03002-05, 07003-08 — size cannot be verified; confirm in field / PMS.
- **Partially legible marks:** slop-tank new line number "2"-DA-2[6]81-A2" (04001-09); JT-skid LCV valve sizes 1⁄4" vs 3⁄4" (07002-11/-14); 8.3 printed bleed size (02004-03); "Has to move" script (03004-06); anti-foam "inst..." (08005-04). Flagged rather than guessed.
- **Pressure class / material for nearly all added small-bore items:** not stated on P&IDs — order per governing line class; Verify in Piping Material Specification.
- **Set-pressure confirmations outstanding** (field "Confirm" marks): PSV-1001, PSV-1003, PSV T-1, PSV V-1, PSV V-2, PSV F-1, PSV T-2, PSV P-3C, PSV-1240A, PSV-3200A, PSV-45001A — verify against nameplates/certs and record.
- **Bore confirmations:** RO T-1 (1.5"), FE F-1 (2.0"), FE P-3 (3.5") — confirm against datasheets.

8 QC / Verification Statement

- All 40 sheets reviewed at high magnification (150 DPI full-sheet + 2-4x zoom crops; 400 DPI re-renders for hard-to-read marks).
- Second pass completed: automated red/pink cluster detection on every sheet compared against the transcribed items — every detected cluster maps to a punch item (the only unaccounted red region per sheet is the printed MPLX logo). Pages 13-14, 23, 28, 33, 36-40 additionally re-verified crop-by-crop.
- Cross-sheet consistency checks: paired markups reconciled (01001↔07004 drain line; 01003↔02001 line 2"-LH-0009; 02005↔02006 drain header 2-DR-0100; 02015C↔02015D blanket gas; 07002↔07005 methanol tie-in; 02013↔02115 mirrored T.O. trains).
- No specifications invented — unknowns are marked "Verify in Piping Material Specification" or listed in Section 7.
- Illegible or ambiguous text is transcribed with [brackets] and flagged, never silently guessed.

Prepared by Claude (construction QA/QC redline review) from the field-marked P&ID set "hayfield_red_lines_and_punch_list_.pdf" (40 sheets), 2026-08-09. Punch item keys reference sheet + item (e.g. 02011-02). This document consolidates field redlines for construction action, procurement, and engineering disposition; it does not supersede issued-for-construction drawings.